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SUPERSEDING
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FEDERAL SPECIFICATION

INK, DRAWING, WATERPROOF, BLACK (FOR USE ON WATER-REPELLENT AND PLASTIC DRAFTING SURFACES)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for black, waterproof drawing ink for use with art and drafting instruments on water-repellent tracing cloth and plastic drafting surfaces.

1.1.1 Definition. For purposes of this specification, "plastic drafting surfaces" are defined as surfaces on which drawings or tracings can be made with a 3H-graphite-lead pencil. "Plastic drafting surfaces" include processed polyester, vinyl, and cellulose acetate film.

1.2 Classification.

1.2.1 Type. The drawing ink shall be of but one type (see 1.1).

1.2.2 Sizes (individual bottles of ink)(see 5.1.1.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

L-P-390 - Plastic, Molding, and Extrusion Material, Polyethylene and Copolymers (Low, Medium, and High Density).
L-P-519 - Plastic Sheet, Tracing, Glazed and Matte Finish.
CCC-C-531 - Cloth, Tracing.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-665 - Boxes, Paperboard, Metal Edged and Components.

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

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(Federal Government activities may obtain copies of the Federal Specifications, Standards, and Handbooks, and the Index of Federal Specifications and Standards, from established distribution points in their agencies.)

Military Specification:

MIL-L-10547 - Liners, Case and Sheet, Overwrap, Water-Vaporproof or Waterproof, Flexible.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, NW., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Composition and stability of ink. The ink shall be composed of finely divided carbon in a stable state of suspension in a liquid medium containing suitable water-proofing agents. The ink shall not show any coagulation, granulation, or precipitation under working and storage conditions. Such evidence of instability shall not be present when the ink is received nor when the sample ink is subjected to the stability tests specified in 4.4.1.

3.2 Performance. The ink, as received and after exposure as specified in 4.4.1, shall have the following performance characteristics when tested as specified in 4.4.2. The ink shall not dry in the pen. The ink shall flow freely and produce lines of uniform width on the specified tracing cloth and plastic surfaces. The ink shall not penetrate the surface of the matte side of the tracing cloth to the extent that a complete erasure cannot be made. The color shall be jet black and the ink shall have a high degree of opacity and a low degree of gloss. The ink shall show no sign of abrasion when the test marking is cleaned with an eraser of the "art gum" type, as specified in 4.4.2.4. The ink shall be non-etching and shall meet the test specified in 4.4.2.5 without causing decomposition of the technical fountain pen tip.

3.3 Resistance to solvents. The ink, when thoroughly dried, shall be insoluble in cold water and in organic solvents, hexane, and 1,1,1 - trichloroethane, when tested as specified in 4.4.3.1 and 4.4.3.2.

3.4 Specific gravity. The ink shall have a specific gravity at 25°C. of not less than 1.055 when determined as specified in 4.4.5.

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3.5 Identification marking. Each bottle of ink shall be clearly and legibly lithographed or marked with water-resistant ink, or furnished with a label printed with water-resistant ink on a good grade of printing paper and firmly affixed to the bottle with a water-resistant adhesive. The markings on the bottle and its individual box shall clearly identify the product and its manufacturer, and be in accordance with standard commercial practice (see 5.3).

3.6 Workmanship. The ink shall be a smooth, homogeneous, black liquid, free from lumps, strings, and objectionable odor. Labels shall be clearly printed and neatly and securely affixed to bottles. Ink containers shall show no evidence of spillage or leakage.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements (see 6.2).

4.2 Sampling and inspection for acceptance. Sampling and inspection shall be performed in accordance with the provisions of MIL-STD-105, except where otherwise indicated hereinafter. For the purposes of sampling, an inspection lot for acceptance, inspection, and tests shall consist of all material and all bottles of ink of the same size, submitted for inspection and delivery at one time.

4.3 Inspection.

4.3.1 Inspection of materials and components. The supplier is responsible, in accordance with 4.1, for insuring that components and materials used were manufactured, tested, and inspected in accordance with the specified requirements of subsidiary specifications and standards, and in accordance with this specification, as applicable.

4.3.2 Inspection of end item and preparation for delivery.

4.3.2.1 Examination of the end item. The end item shall be examined in accordance with the classification of defects listed for each of the applicable examinations.

4.3.2.1.1 Examination of individual containers of ink for defects in appearance, design, and workmanship. An examination shall be made as indicated in table I. The sample unit for this examination shall be one filled glass bottle or polyethylene container, as applicable. The inspection and acceptable quality levels (AQL's) shall be as specified in 4.3.3 with the lot size expressed in units of individual containers of ink.

TABLE I. Visual examination (end item)

Examine	Defects
Appearance	Not homogeneous. Contains lumps or strings. Stray spilling of ink or leakage on outside of bottle.
Design	Container and closure not as specified.
Fill of individual container (except average contents)	Empty bottle, or filled to extent that bottle may burst or leak in hot weather due to lack of liquid expansion space.

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4.3.2.1.2 Examination of individual containers for average contents. The sample unit for this examination shall be one filled glass bottle or polyethylene container, as applicable. The inspection level shall be as specified in 4.3.3 with the lot size expressed in units of individual containers of ink. The average contents shall be not less than that specified.

4.3.2.2 Examination of preparation for delivery. An examination shall be made to determine compliance with the packaging, packing, and marking requirements of section 5 and contract or order. Defects shall be scored in accordance with table II. For examination of contents, the sample unit shall be one shipping container fully prepared for delivery, selected just prior to the closing operations. Defects of closure specified shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-4 with an AQL of 4.0 defects per 100 units.

4.3.3 Inspection levels and AQL's. The inspection levels and AQL's for examination of the end item shall be as follows:

Specification reference	Inspection level	AQL
4.3.2.1.1	S-4	2.5
4.3.2.1.2	S-2 (See 4.3.2.2)	-

TABLE II. Examination of preparation for delivery

Examine	Defects
Markings (unit, intermediate, and exterior containers, as applicable)	Incorrect; omitted; illegible; improper size, location, sequence, or method of application; omission "THIS SIDE UP" and other warnings, as applicable.
Materials	Wrong type, class, or other category of required specification. Material missing, damaged, or defective, affecting serviceability.
Workmanship	Wrong or careless means in placement and use of labeling and packaging and packing materials such as loose or partially attached labels; flaps of containers incompletely closed; loose or overtight straps, dangerous strap ends, protruding nails, and smeared adhesive.
Contents	Quantities of unit and intermediate containers not as specified. Wrong arrangement; bottle on side or upside down.
Closure of shipping containers	Not as specified.
Gross weight of shipping containers	In excess of specified weight. In excess of box limitation shown on boxmaker's certificate.

4.3.4 Testing of ink. Each lot or batch of drawing ink shall be tested in accordance with table III. Each sample unit shall meet all requirements. One sample unit shall consist of 1 pint of ink from one or more bottles of ink kept at ordinary temperature. Another sample unit of 1 pint shall consist of ink from one or more bottles after exposure is made in the bottles as specified in 4.4.1.2.

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4.3.5 Testing of polyethylene containers for permeability. Five 1-ounce polyethylene containers and five 1-ounce glass bottles, filled with ink from the same lot, shall be tested in accordance with 4.4.4. The average weight loss of the polyethylene containers shall not exceed the maximum loss specified in 5.1.1.1 when the loss is computed in accordance with the specified test method.

4.4 Tests.

TABLE III. Instructions for testing of ink

Characteristic	Specification reference: Requirement	Specification reference: Test method	Requirements applicable to individual unit	Number determinations per unit	Results reported as pass or fail	Testing level
Stability.	3.1	4.4.1.2	X	1	X	Two sample units (see 4.3.4)
Performance (before and after exposure).	3.2	4.4.2				
Ink flow and uniformity of line width on:						
Tracing cloth.		4.4.2.1	X	1	X	
Plastic film.		4.4.2.1	X	1	X	
Penetration.		4.4.2.2	X	1	X	
Color, opacity, and gloss.		4.4.2.3	X	1	X	
Abrasion resistance.		4.4.2.4	X	1	X	
Non-etching characteristic.		4.4.2.5	X	1	X	
Resistance to solvents.	3.3	4.4.3				
Water.		4.4.3.1	X	1	X	
Hexane.		4.4.3.2	X	1	X	
1,1,1 - trichloroethane.		-do-	X	1	X	
Specific gravity at 25°C.	3.4	4.4.5	X	1	X	

4.4.1 Stability of ink.

4.4.1.1 Original stability. Examine the sample of ink in its original container for precipitation by inserting a glass rod. If present, invert bottle and shake vigorously for 1 minute. Examine again for sediment and coagulation. Place a few drops of ink between glass plates, rub plates together, and then examine for granulation.

4.4.1.2 Stability to temperature changes. Samples of ink in their original closed containers shall be exposed, each container to both of the temperature conditions listed below:

(a) At 125° ± 2°F. for 4 hours.
(b) At 32° ± 2°F. for 4 hours.

The sample ink, after exposure to the above specified temperatures and adjustment to room temperature, shall be examined for coagulation, granulation, and precipitation when tested as specified in 4.4.1.1.

4.4.2 Performance. The performance tests shall be made with clean and properly sharpened drawing pens and the ink shall be observed for drying out in the pens during these tests.

4.4.2.1 Ink flow and uniformity of line width.

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4.4.2.1.1 Test 1. Set a drawing pen at an opening of 0.003 ± 0.001 inch and fill it with sufficient ink to draw at least five lines, each 6 inches long. These lines shall be drawn parallel at intervals of 1/2 minute on a matte side of white tracing cloth conforming to type III of CCC-C-531, and on matte-finish plastic sheet (film) conforming to type I of L-P-519. The ink shall be considered unsatisfactory if it is not possible to draw five clean, sharp, unbroken lines of uniform width. The ruler used shall have a square edge. The pen shall release the ink immediately upon being drawn along the straight edge. The pen shall be lifted vertically from the drawing material at the end of each 6-inch line.

4.4.2.1.2 Test 2. Set a detail drawing pen with an opening of 0.025 ± 0.001 inch and draw 10 parallel lines 2 inches long, 1/4 inch apart, on the specified plastic film, and on both sides of the specified tracing cloth. (The tracing cloth shall be matte finish on both sides.) After these lines are allowed to dry for 10 minutes, 10 parallel lines of the same width and length shall be drawn at right angles across each of the sets previously drawn. The pen shall contain approximately the same volume of ink each time it is charged. There shall be no change in the setting of the pen at any time during the progress of the test. The ruler shall be as specified in 4.4.2.1.1. At the completion of the test, there shall be no noticeable difference in the thickness of the lines on any surface; the intersecting lines shall be free from distortion or blots caused by the ink spreading under the ruling edge. The pen shall release ink immediately upon being drawn along the straight edge. The pen shall be lifted vertically from the drawing material at the end of each 2 inch line.

4.4.2.2 Penetration. A sample of the markings prepared in 4.4.2.1.2 shall be rubbed with an eraser of the "Van Dyke" No. 6500 type or equal. This test shall be conducted on a matte side of the tracing cloth and over a 1 inch portion of the markings. The ink markings shall be completely erasable.

4.4.2.3 Color, opacity, and gloss. Draw two lines, 12 inches long, on the specified plastic film, with a technical fountain pen, using ink obtained from two different samples of the same lot. After the lines are dry, apply another coat on the second line with the second ink. Examine the middle portion of each 12 inch line for intensity of color (blackness), opacity, and degree of gloss.

4.4.2.4 Abrasion resistance. A sample of markings, prepared as specified in 4.4.2.3, shall be rubbed vigorously with an eraser of the "art gum" type or equal. There shall be no perceptible evidence of ink removal.

4.4.2.5 Non-etching characteristic of ink. Following test specified in 4.4.2.3, immerse the tip of the technical fountain pen in sample ink used in the test and allow the pen tip to remain immersed for 48 hours. After the specified period of immersion, withdraw the pen from the ink and examine for compliance with the applicable requirement of 3.2.

4.4.3 Insolubility in water and resistance to organic solvents tests. Draw two parallel lines, approximately 0.025-inch wide, 1/4 inch apart, and 8 inches long on the specified tracing cloth and plastic film and mark off into two equal parts. After these have been allowed to dry for at least 1 hour, each part shall be subjected to the following tests:

4.4.3.1 Insolubility in water. Place four drops of water (70° - 80°F.) and spread it over both lines for a length of about 1 inch, without rubbing or friction. Allow the water to evaporate naturally. There shall be no dissolution of the ink and no spreading of the lines.

4.4.3.2 Resistance to organic solvents. Repeat the above operation using hexane and 1,1,1 - trichloroethane on separate areas. Additional quantities of each should be added in order to keep the surface wet at least 5 minutes. Allow the solvents to evaporate naturally. Observe for evidence of dissolution of the ink, spreading of the lines, or reduction of intensity of color (blackness).

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4.4.4 Permeability test. Five ink-filled 1-ounce polyethylene containers of each inspection sample and five ink-filled 1-ounce glass bottles of the same ink representing control samples shall be sealed with paraffin wax along the edge of the stoppers, and individually weighed at room temperature. The ink and containers shall then be placed inside a temperature-controlled test chamber and maintained at a temperature of 100° ± 1°F. for 28 consecutive days. After this period of time, the containers shall be removed, allowed to cool at room temperature for 1 day, and then reweighed. The average weight loss of the test samples minus the average weight loss of the control samples shall be the average weight loss of the test samples due to permeation. If the average weight loss is greater than 0.4 gram per 1-ounce container, the lot shall be rejected (see 5.1.1.1).

4.4.5 Specific gravity. The specific gravity of the ink shall be determined at 25°C. by using a pycnometer.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit package. One, two, four, six, eight, sixteen, and thirty-two ounces of ink shall be packaged in 1-, 2-, 4-, 6-, 8-, 16-, and 32-ounce bottles, respectively, and shall be furnished as specified (see 6.2). The bottles shall be fabricated from commercial grade glass or from polyethylene conforming to type I, grade 1, of L-P-390. If a polyethylene container is used, the average weight loss due to permeation of the ink through the container shall not exceed 0.4 gram per 1-ounce container when tested as specified in 4.4.4. The shape of 1-ounce bottles shall be such that they can be inserted in the ink bottle holders commercially available. Each 1-ounce bottle shall be provided with a leakproof stopper and ink applicator in accordance with standard commercial practice. The stopper and applicator shall not be affected by the ink, nor shall the ink be affected by the stopper and applicator. The applicator shall extend down to within 1/8 inch of the bottom of the bottle when the stopper is tightly in place. Each 2-, 4-, 6-, 8-, 16-, and 32-ounce bottle shall be provided with a plastic continuous thread screw cap closure (see 5.1.1.1.1). They shall be of adequate weight sidewall so that an average of not more than 10 out of 30 caps will be broken in application by capping machines at 30 inch-pounds torque and no individual cap will break at less than 15 inch-pounds. All screw caps shall have a non-absorbent resilient lining or a lining backed by resilient material which will remain in the cap. An inner seal shall be used to cover and adhere to the mouth of the 2-, 4-, 6-, 8-, 16-, and 32-ounce bottles which will prevent loss of the contents should the cap become loosened. Caps shall be tightened to prevent leakage. Each glass bottle and each 1-ounce plastic bottle shall be packaged in a snug-fitting box conforming to style I or II of PPP-B-566. Boxes containing bottles of 2-, 4-, or 6-ounce size shall have tuck ends of a lock type as illustrated in figure 2 of PPP-B-566 for the style of box specified. In addition, each box containing 16- or 32-ounce glass bottles shall be fitted with a single-faced corrugated liner. Each individual box containing 8-, 16-, or 32-ounce bottles shall be sealed with tape as specified in the appendix to the box specification.

5.1.1.1.1 Alternate closure. Alternate continuous-thread screw cap closure for the 2-, 4-, 6-, 8-, 16-, and 32-ounce size bottles, of pour-out type or other design in accordance with standard commercial practice, will be acceptable provided the alternate closure meets the quality established by this specification and is at least equally effective to that described above for the purpose intended.

5.1.1.2 Intermediate package. Ink, unit packaged as specified in 5.1.1.1, except for 16-ounce and 32-ounce bottles, shall be intermediate packaged in quantities specified in table IV in a snug-fitting box as specified in 5.1.1.1; or shrink wrapped. Sixteen-ounce and 32-ounce bottles shall be intermediate packaged in boxes conforming to type CF or SF, class domestic, of PPP-B-636 or PPP-B-665; in quantities as indicated in table IV.

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TABLE IV

Size (Ounces)	Quantity of packages per intermediate package	Quantity of unit packages per shipping container
1	6	144
2	6	144
4	6	144
6	6	96
8	6	96
16	6	48
32	6	24

5.1.2 Level C. Unless otherwise specified, each bottle of ink as specified in 5.1.1.1 shall be packaged in accordance with standard commercial practice.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Bottles, packaged as specified in 5.1, shall be packed in quantities as shown in table IV in a snug-fitting fiberboard shipping container conforming to class weather-resistant, V3c or V3s, style CSSC-L of PPP-B-636. The inside of each fiberboard shipping container shall be fitted with a taped one-piece double-faced corrugated liner having a minimum bursting strength of 275 p.s.i. Each shipping container shall be provided with a type I or II, grade B or C, class 2 sealed case liner conforming to MIL-L-10547, placed between the corrugated liner and the inner surfaces of the container. Each shipping container shall be closed and strapped in accordance with the appendix to the container specification.

5.2.2 Level B. Bottles, packaged as specified in 5.1, shall be packed in quantities as specified in table IV in a snug-fitting fiberboard shipping container conforming to type CF or SF, class domestic, style RSC, of PPP-B-636, except that the minimum bursting strength of the fiberboard shall be 200 p.s.i. and each box shall be furnished with top and bottom pads of minimum 200 p.s.i. test, double-faced "C" flute fiberboard. Box closure shall be in accordance with method II of the appendix to the box specification.

5.2.3 Level C. Unless otherwise specified, individual containers, packaged as specified in 5.1, shall be packed in shipping containers of the type, size, and kind commonly used in commercial practice, in a manner that will insure acceptance by common carriers and safe delivery at destination. Shipping containers shall comply with National Motor Freight Classification rules or the Uniform Freight Classification rules.

5.3 Marking. (See 3.5.)

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, intermediate packages and shipping containers shall be marked in accordance with Fed. Std. No. 123. Unit packages shall be marked in accordance with the manufacturer's commercial practice.

5.3.2 Military agencies. In addition to any special marking required by the contract or order, intermediate packages and shipping containers shall be marked in accordance with MIL-STD-129. Unit packages shall be marked in accordance with the manufacturer's commercial practice.

5.3.3 Special marking. Each shipping container containing glass bottles shall bear "FRAGILE" markings. All shipping containers shall bear arrows and the word "UP" on at least one side and one end and the words "TOP" and "THIS SIDE UP" on the top surface.

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6. NOTES

6.1 Intended use. The black waterproof drawing ink covered herein is intended primarily for use with artists' and draftsmen's instruments on water-repellent tracing cloth for ink work and plastic drafting surfaces, as specified herein.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Data, as required; inspection responsibility if other than as specified (see 4.1).
(c) Size of unit package (individual ink container)(see 5.1.1.1).
(d) Level of packaging, level of packing, and any special marking required (see 5.1, 5.2, and 5.3).

6.3 Definitions. For purposes of this specification, "coagulation," "granulation," and "precipitation" are defined as follows:

Coagulation:
The result of gelation.

Granulation:
Hard particles which can be detected by placing a few milliliters of ink between glass plates and then rubbing the plates together.

Precipitation:
Precipitates are defined as particles of solids which fall out of solution or suspension and which cannot be redispersed or suspended by either inverting or shaking bottle by hand for 1 minute.

User Interest:

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Preparing Activity:

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